

Standalone SMITH / SONIC manual

SMITH/SONIC standalone distribution

1. Scope

Standalone SMITH converts a Cartesian molecular description and its perceived molecular state into a rank-complete SONIC internal-coordinate contract. It is the small, reproducible interface associated with the SMITH manuscript, not a distribution of the complete MATRIX framework.

ORACLE and SMITH have distinct roles. ORACLE performs continuous molecular perception, including topology and cycles, symmetry and atom equivalence, effective atomic numbers, synthons, and interaction centres. It develops the ideas introduced in PROXIMA and is in final testing; it will be released and described separately when ready. SMITH consumes a frozen ORACLE state and constructs coordinate candidates, protected special coordinates, rank reduction, symmetry adaptation, analytic Wilson rows, and the serialized SONIC contract.

2. Requirements and installation

Use Python 3.11 or newer and Git. Both SMITH and its pinned MATRIX dependency are private during the collaborator test. First accept both GitHub invitations and authenticate Git on the command line:

```
gh auth login
gh auth setup-git
git clone https://github.com/yogibubu/Smith.git
cd Smith
git switch agent/oracle-boundary-standalone-smith
python3 -m venv .venv
source .venv/bin/activate
python -m pip install --upgrade pip
python -m pip install ./standalone
```

On Windows PowerShell, activate the environment with `.venv\Scripts\Activate.ps1`. The branch-switch command will no longer be needed after the draft branch is merged.

3. Command summary

```
smith-sonic --version
smith-sonic example NAME [OUTPUT]
smith-sonic build INPUT [OUTPUT] [--require-oracle-state]
smith-sonic inspect FILE
```

`example` runs an input installed with the package. `build` accepts either a plain extended XYZ input or an ORACLE-enriched `xyzin`. `inspect` reports the presence of the frozen state, GIC section, and provenance profile.

Every `build` or `example` command writes three files with the same stem:

- `.xyzin`: frozen ORACLE/SMITH/SONIC contract;
- `.smith.out`: readable coordinate report with rank, families, protected rows, values, units, irreps, active/frozen status, and primitive coefficients;
- `.g16.gjf`: commercial Gaussian 16 `ReadAllGIC` optimization input.

The Gaussian 16 profile is enabled by default. It uses Gaussian-compatible improper dihedrals and marks every non-totally symmetric coordinate as `Frozen`; totally symmetric coordinates remain active. Fragment and interaction-center helper functions are serialized as `Inactive` definitions.

For a production state prepared and validated by ORACLE, require the boundary explicitly:

```
smith-sonic build molecule.oracle.xyzin molecule.sonic.xyzin --require-oracle-state
```

This option refuses an input missing any of `VALIDATION`, `TOPOLOGY`, `SYNTHONS`, or `SYMMETRY`.

4. Input profiles

Frozen ORACLE state

This is the production interface. SMITH preserves the molecular perception sections and adds the SONIC construction. The output provenance contains `PERCEPTION_PROFILE` `ORACLE_STATE`.

Reduced ORACLE convenience profile

A plain or extended XYZ is sufficient for small reproducibility examples. The packaged path performs a reduced perception pass and records `PERCEPTION_PROFILE` `REDUCED_ORACLE`. For disconnected components it also constructs fragment definitions and six intermolecular translation/orientation coordinates. This profile is deliberately limited and is not a substitute for the forthcoming ORACLE application.

Extended-XYZ directives follow the Cartesian block, for example:

```
#SMITH
fragment_mode = special-coordinates
symmetrize = false
sycart = false
improper_dihedrals = false
```

The available advanced controls include `symmetry_group`, `xh_stretch_policy`, `local_xh_bonds`, and `local_xh_classes`. Omit them to use the pinned SMITH defaults.

5. Packaged examples

Run the complete set with:

```
smith-sonic example water water.xyzin
smith-sonic example norbornane norbornane.xyzin
smith-sonic example formic-acid-water formic-acid-water.xyzin
smith-sonic example eta3-allyl-palladium eta3-allyl-palladium.xyzin
```

Expected results are:

Example	Input profile	Target rank	Purpose
water	reduced	3	minimal nonlinear molecule
norbornane	reduced	51	bridged and cyclic topology
formic-acid-water	reduced + fragments	18	non-covalent two-fragment contract
eta3-allyl-palladium	frozen ORACLE state	24	protected metal-to- η^3 -centre coordinate

The formic-acid-water output must contain three `FRAG_TRANSLATION` and three `FRAG_ORIENTATION` primitives. The η^3 probe must retain an `ETA3_CENTER` over the three allyl carbon atoms and create a protected `CENTER_ATOM_DISTANCE` from Pd to that centre.

The η^3 geometry is an idealized interface test, not an optimized or computed chemical benchmark. Its interaction centre is supplied explicitly in a frozen ORACLE-state fixture. It tests whether SMITH consumes the correct contract; it does not claim that the reduced profile can perceive η^3 bonding.

The advanced generated artifacts are retained under `standalone/examples/`:

- `formic_acid_water.xyzin`, `.smith.out`, and `.g16.gjf`;
- `eta3_allyl_palladium.xyzin`, `.smith.out`, and `.g16.gjf`.

6. Reading an output

The most relevant sections are:

- `TOPOLOGY`, `SYNTHONS`, `SYMMETRY`, and `INTERACTION_CENTERS`: perceived

molecular state consumed by SMITH;

- `GIC`: build options, target dimension, primitive candidates, protected

special coordinates, and the selected generalized internal coordinates;

- `SMITH_PROVENANCE`: package version, pinned MATRIX revision, and perception

and fragment profiles.

For a nonlinear system of N atoms without disconnected fragments, the usual target is $3N - 6$. Always use the serialized `TARGET_RANK` and reported final rank when special fragment or centre coordinates are present.

7. Verification and problem reports

From the repository root, run:

```
python -m unittest discover -s standalone/tests -v
```

With the full MATRIX checkout on `PYTHONPATH`, the same suite also runs the MORPHEUS/LINK consumer test. It requires direct MORPHEUS parsing of each `xyzin` and construction of LINK SONIC direction matrices with shapes 18×24 and 24×30 .

When reporting a problem, include the operating system and architecture, Python version, exact command, complete terminal output, input file, and generated `xyzin` if one was produced. Do not include credentials or private tokens. During this test phase, open an issue in the private repository or send the report directly to the project owner.

Execution of Gaussian jobs, full optimization workflows, and the full ORACLE validation surface remain outside this reduced package. The package generates the Gaussian 16 input but does not launch the external executable.